

Problem

A Wheatstone bridge has $R_1 = R_3 = 2 \text{ k}\Omega$. R_2 is adjusted until no current flows through the galvanometer. At that point, $R_2 = 6.3 \text{ k}\Omega$. What is the value of the unknown resistance?

Step-by-step solution

Step 1 of 1

Refer to Figure 4.63 in the text book.

Calculate the value of the unknown resistance, R_x .

$$R_x = \frac{R_3 R_2}{R_1}$$

Substitute $6.3 \text{ k}\Omega$ for R_2 , $2 \text{ k}\Omega$ for R_1 and $2 \text{ k}\Omega$ for R_3 in the equation.

$$\begin{aligned} R_x &= \frac{(2 \text{ k})(6.3 \text{ k})}{2 \text{ k}} \\ &= 6.3 \text{ k}\Omega \end{aligned}$$

Therefore, the value of the unknown resistance R_x is $6.3 \text{ k}\Omega$.